

PROJECT I

SUPPLY CHAIN GREENHOUSE GAS EMISSION FACTOR ANALYSIS – 2024

1. Project Overview

This project focuses on analyzing supply chain greenhouse gas emission factors across different industries using NAICS-based environmental data. The analysis helps identify high emission industries, compare emission intensity across different industrial sectors, and understand the environmental impact of supply chain activities. By using Excel and Power BI, the project converts raw environmental data into meaningful insights through data cleaning, visualization, and KPI analysis.

2. Objectives

The key goals of this project are:

- Analyze overall greenhouse gas emission performance across industries
 - Identify industries with the highest emission margins
 - Categorize industries into High, Medium, and Low emission groups
 - Compare industry margins with emission contribution
 - Analyze the relationship between margins and emission levels
 - Identify industry-wise emission contribution percentages using visual analytics.
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3. Data Sources

- **Dataset Source:** Data.gov
 - **Domain:** Supply Chain & Environmental Sustainability
 - **Timeline:** 2024
 - **Data Size:** 1016 Rows and 8 columns
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4. Data Cleaning (Preprocessing)

The following preprocessing steps were performed to ensure data quality:

- Removed duplicate records from the dataset
- Corrected inconsistent data types
- Renamed columns for better readability and understanding
- Filtered invalid and irrelevant records
- Categorized emission levels into High, Medium, and Low groups.

- Checked for outliers and abnormal emission values. The dataset contained no null or blank values.

Tools & Technologies Used:

- Power BI
- Excel / CSV Dataset
- Data Cleaning
- Data Visualization
- DAX Functions

5. Data Transformation

To enhance the analysis and improve data quality, the following transformations were applied to the dataset:

- Formatted numerical columns to maintain consistent decimal values and data types.
- Created calculated columns such as:
 - Emission Difference
 - Margin Contribution %
 - Emission Intensity Category
- Categorized industries into **Low**, **Medium** and **High** emission groups based on emission intensity values.
- Generated statistical summary measures including average, minimum, maximum, range, and standard deviation for emission analysis.
- Created Pivot Tables to analyze industry-wise average emission factors and emission category distribution.
- Developed KPI measures in Power BI to evaluate:
 - Average Emission Factor
 - Maximum and Minimum Emission
 - High Emission Industry Count
 - Average Margin Contribution %
 - Total Industries
- Applied sorting, filtering, and conditional formatting techniques to highlight high-emission industries and improve data visualization.

STATISTICS SUMMARY	
<i>Statics Summary of SCEF with margins</i>	
Mean	0.281897638
Median	0.173
Mode	0.111
Standard Deviation	0.321416635
Sample Variance	0.103308654
Kurtosis	30.29919653
Skewness	4.236073543
Range	3.895
Minimum	0.029
Maximum	3.924
Sum	286.408
Count	1016

➤ **Major Pivot Tables:**

2 INDUSTRY vs EMISSION	
Row Labels	Average of With Margins
All Other Pipeline Transportation	1.62
Beef Cattle Ranching and Farming	2.89
Cattle Feedlots	2.89
Cement Manufacturing	3.92
Dairy Cattle and Milk Production	1.72
Dual-Purpose Cattle Ranching and Farming	2.89
Gypsum Product Manufacturing	1.62
Lime Manufacturing	1.62
Pipeline Transportation of Crude Oil	1.62
Pipeline Transportation of Natural Gas	1.62
Pipeline Transportation of Refined Petroleum Products	1.62
Grand Total	2.19

INDUSTRY Vs MARGIN CONTRIBUTION

Row Labels	Average of Margin Contribution
All Other Publishers	51%
Apparel Accessories and Other Apparel Manufacturing	50%
Cut and Sew Apparel Contractors	50%
Directory and Mailing List Publishers	51%
Greeting Card Publishers	51%
Hosiery and Sock Mills	50%
Lawn and Garden Tractor and Home Lawn and Garden Equipment M	57%
Men's and Boys' Cut and Sew Apparel Manufacturing	50%
Other Apparel Knitting Mills	50%
Other Communications Equipment Manufacturing	53%
Other Cut and Sew Apparel Manufacturing	50%
Pharmaceutical Preparation Manufacturing	54%
Primary Battery Manufacturing	55%
Software Publishers	56%
Women's, Girls', and Infants' Cut and Sew Apparel Manufacturing	50%
Grand Total	52%

6. Exploratory Data Analysis (EDA) & Visualization

Exploratory Data Analysis (EDA) was performed to understand emission patterns, compare industry-wise environmental impact, and identify high-emission sectors within the supply chain dataset. Statistical analysis techniques were used to summarize emission factors and evaluate data distribution across industries.

- Performed statistical analysis using measures such as mean, median, minimum, maximum, range, and standard deviation to understand overall emission intensity.
- Conducted univariate analysis through emission distribution visuals and category-based analysis to examine the behavior of individual variables.
- Performed bivariate analysis by comparing emission factors with and without supply chain margins across industries using clustered charts and comparative visuals.
- Conducted multivariate analysis by analyzing the relationship between industry type, emission category, margin contribution, and total emission intensity using interactive charts and tables.

- Built an interactive dashboard in Power BI consisting of KPI cards, Bar charts, Column charts, Pie charts, Scatter plots, Matrix and Tables for effective environmental impact analysis.

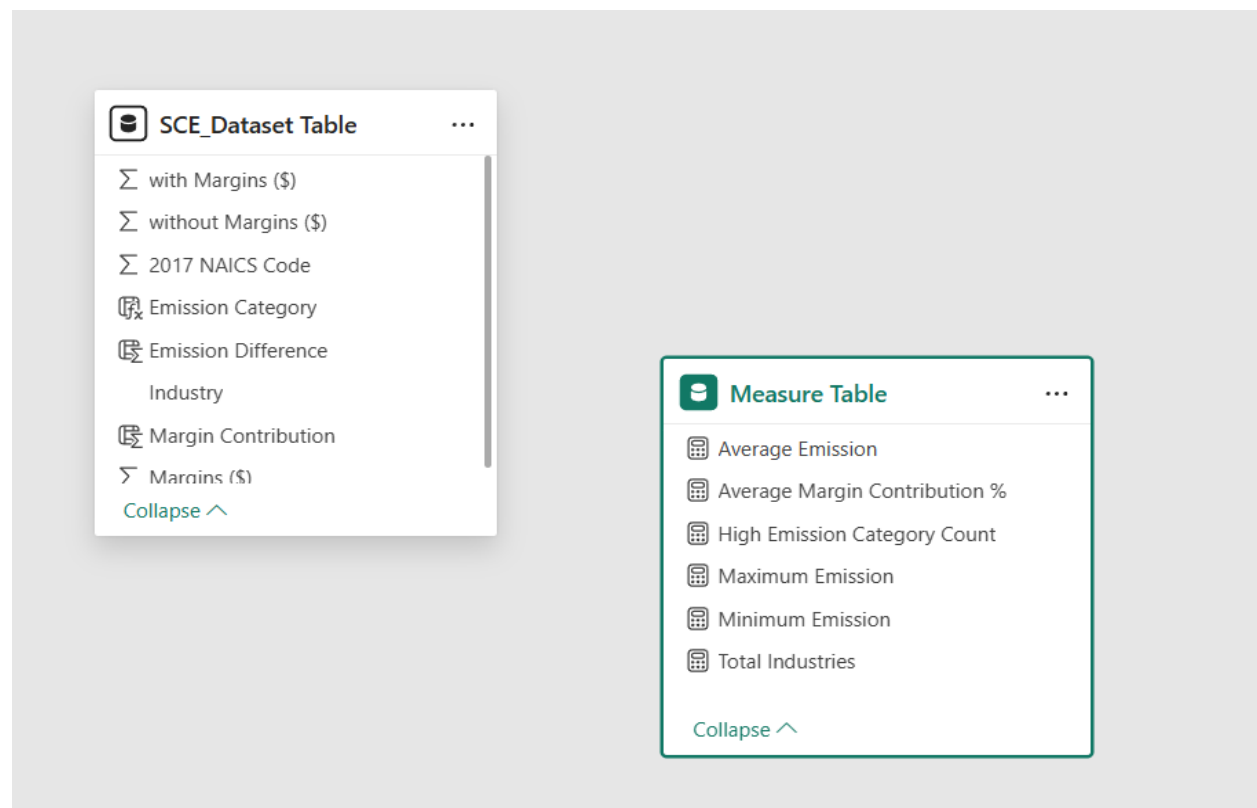
Features Included:

- Interactive visuals for industry-wise emission comparison
- Filters and slicers for dynamic data exploration
- Comparative charts for margin and emission analysis
- Drill-down and detailed industry-level insights through matrix and visual interactions.

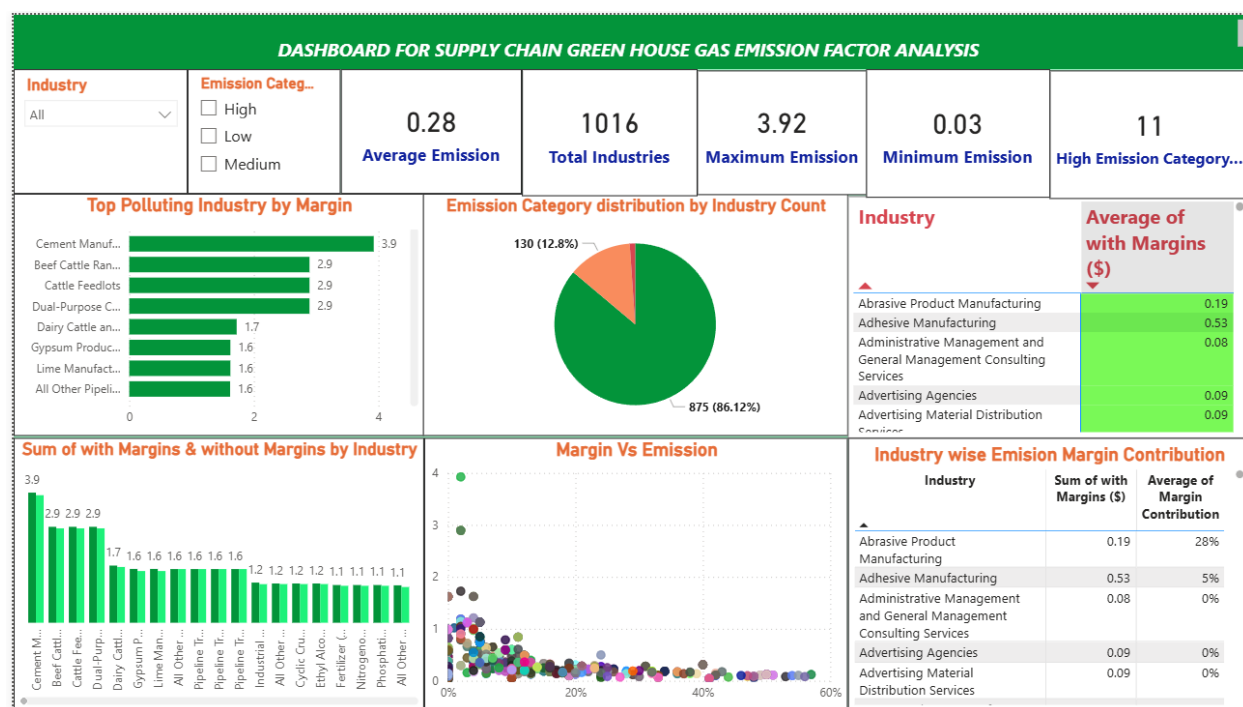
➤ Data Modelling

The project uses a denormalized data model consisting of a single consolidated table containing all industry-wise greenhouse gas emission information. Since the dataset does not contain multiple related tables, no table relationships were created in the Power BI model.

The denormalized structure simplifies data analysis, improves visualization performance, and enables direct KPI and dashboard creation without complex relationship management.



DASHBOARD:



7. Key Insights

OBJECTIVE 1: Identify Top Polluting Industries

Visual: Bar Chart – Top Polluting Industry by Margin

- **Descriptive Analytics:** Cement Manufacturing recorded the highest emission margin of **3.924**, while Beef Cattle Ranching, Cattle Feedlots, and Dual-Purpose Cattle Ranching recorded **2.893** emission margins.
- **Diagnostic Analytics:** High emission margins are caused by energy-intensive manufacturing processes and methane emissions generated from livestock-related industries.
- **Predictive Analytics:** If current industrial operations continue, cement and cattle-related industries are likely to remain the leading greenhouse gas emission contributors in future years.
- **Prescriptive Analytics:** Implementing low-carbon manufacturing methods and sustainable livestock management practices can help reduce emission margins by **10–15%** within the next operational year.

OBJECTIVE 2: Analyze Emission Category Distribution

Visual: Pie Chart – Emission Category Distribution by Industry Count

- **Descriptive Analytics:** The Low Emission category contains **875 industries (86.12%)**, Medium Emission includes **130 industries (12.8%)**, and High Emission consists of only **11 industries (1.08%)**.
- **Diagnostic Analytics:** Most industries fall under the Low Emission category because they involve moderate operational activities, whereas only a few industries perform highly carbon-intensive processes.
- **Predictive Analytics:** Low Emission industries are expected to maintain stable environmental performance, while Medium Emission industries may gradually increase with industrial expansion.
- **Prescriptive Analytics:** Medium and High Emission industries should adopt renewable energy usage, emission monitoring systems, and sustainable operational practices to reduce emission contribution by **8–10%** in the coming years.

OBJECTIVE 3: Compare Industry Margins with Emission Contribution

Visual: Column Chart – Sum of with Margins & without Margins by Industry

- **Descriptive Analytics:** Cement Manufacturing recorded the highest emission contribution with margins at **3.924** and without margins at **3.846**, while cattle-related industries such as Beef Cattle Ranching and Cattle Feedlots recorded values of **2.893** and **2.847** respectively.
- **Diagnostic Analytics:** Higher margin contributions are mainly caused by large-scale production activities, transportation operations, and intensive energy consumption within manufacturing and livestock-related industries.
- **Predictive Analytics:** Industries with consistently high margins may continue dominating operational contribution but could face increased environmental compliance costs in future years.
- **Prescriptive Analytics:** Organizations should balance profitability with sustainability by optimizing resource usage and adopting eco-friendly production methods to reduce emission impact by **10–12%**.

OBJECTIVE 4: Analyze Relationship Between Margin and Emission Levels

Visual: Scatter Plot – Margin Vs Emission

- **Descriptive Analytics:** The scatter plot indicates that most industries are concentrated within lower emission ranges between **0.029 and 0.329**, while only a few industries recorded extremely high emission values close to **3.924**.
- **Diagnostic Analytics:** Moderate operational activities and lower resource consumption resulted in lower emissions for most industries, whereas large-scale manufacturing and livestock-related industries contributed to extreme emission values.

- **Predictive Analytics:** If current high-emission operational patterns continue, industries with emission values above **2.0** may face increased environmental risks, sustainability pressure, and operational expenses in future years.
- **Prescriptive Analytics:** Continuous emission monitoring, optimized resource utilization, and sustainable operational planning can help industries reduce high emission occurrences by **10%** over the next business cycle.

OBJECTIVE 5: Evaluate Industry-wise Emission Margin Contribution

Visual: Table – Industry Vs Emission Margin & Contribution

- **Descriptive Analytics:** Adhesive Product Manufacturing recorded an average margin contribution of 0.53, while Abrasive Product Manufacturing and Administrative sectors recorded 0.19 and 0.08 respectively.
- **Diagnostic Analytics:** Manufacturing industries contribute higher emission margins due to machinery usage, transportation processes, and energy-intensive production activities.
- **Predictive Analytics:** Industries with higher contribution percentages are likely to face stricter environmental regulations and sustainability requirements in future operations.
- **Prescriptive Analytics:** Implementing carbon footprint tracking and efficient supply chain management can help reduce emission contribution percentages by 8–10% within the next operational period.

OBJECTIVE 6: Compare Industry wise Margin Contribution Patterns

Visual: Matrix Visual – Industry Contribution Analysis

- **Descriptive Analytics:** The matrix visual compares industry-wise average margins and contribution values, where manufacturing industries showed higher contribution percentages than administrative sectors.
- **Diagnostic Analytics:** Higher contribution patterns occur because manufacturing industries involve larger production activities, transportation processes, and energy-intensive operations.
- **Predictive Analytics:** Industries with higher contribution trends may continue increasing overall greenhouse gas emissions if operational activities expand further.
- **Prescriptive Analytics:** Industries should optimize operational efficiency and implement sustainable production strategies to reduce contribution percentages by **8–10%** within the next business cycle.

Overall Key Insights

- Cement Manufacturing recorded the highest emission margin of **3.924**, making it the major greenhouse gas emission contributor in the dataset.

- The **Low Emission** category dominated with **875 industries (86.12%)**, while only **11 industries (1.08%)** belonged to the High Emission category.
- Manufacturing and livestock-related industries generated higher emissions due to energy-intensive production processes and methane emissions.
- Most industries operated within lower emission ranges, whereas only a few industries showed extremely high emission values close to **4.0**.
- Continued industrial expansion may increase environmental regulations and operational costs.
- Implementing renewable energy usage, sustainable production methods, and carbon monitoring systems can help reduce emission levels by **10–15%** within the next operational year.

8. Conclusion

The project successfully analyzed greenhouse gas emission factors across different industries using interactive dashboards and data visualization techniques. The analysis identified key emission patterns, industry-wise contribution levels, and high-polluting sectors within the supply chain. Data-driven insights helped recommend sustainable operational strategies and environmental improvement measures. Overall, the project demonstrates how business intelligence tools can support effective environmental and supply chain decision-making.